



Patient-Centered Care

KING FAISAL HOSPITAL RWANDA

Clinical Guideline No: G19/2014

**Guideline Title: COMMUNITY
ACQUIRED PNEUMONIA (CAP)
MANAGEMENT**

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INTRODUCTION

Community acquired Pneumonia (CAP) is defined as an acute infection of the pulmonary parenchyma that is acquired outside a hospital or healthcare setting. It typically presents with symptoms of acute infection, accompanied by the presence of an acute infiltrate on a chest radiograph or auscultatory findings (such as altered breath sounds and/or localized rales). To diagnose CAP, at least one systemic feature (either a symptom complex of sweating, fevers, shivers, aches and pains and/or temperature of 38°C or more) must be present, with no alternative explanation for the illness. Community acquired pneumonia remains a disease with high mortality if not recognized timely and correctly managed.

PURPOSE

To provide accurate and update information to care providers who manage CAP cases in the hospital so that diagnosis and appropriate management of CAP are done timely and correctly.

AIM

To improve outcome of patients presenting at King Faisal Hospital with community acquired pneumonia.

RATIONALE

Failure to diagnose CAP timely or administering the wrong antibiotics, can have serious and potentially fatal consequences. This guideline provides recommendations to prevent delay or incorrect treatment, aiming to reduce hospital stay and minimize complications from pneumonia.

SCOPE OF GUIDELINE

This guideline is intended for all healthcare providers managing adult or adolescent patients with community-acquired pneumonia. This can be in the general outpatient department, internal medicine departments and Accident & Emergency department. The guideline applies to all adult or adolescent patients presenting with CAP that are seen in King Faisal Hospital, both inpatients and outpatients.

COMPETENCIES REQUIRED

Every medical doctor working in the mentioned areas where patients with CAP are managed has access to the guidelines and is competent to apply them.

DEFINITION OF KEY TERMS

CAP: Community Acquired Pneumonia

COPD: Chronic Obstructive Pulmonary Disease

CURB-65: scoring system used to assess the severity of pneumonia based on 5 criteria: confusion, urea, respiratory rate, blood pressure, and age

HIV: Human Immunodeficiency Virus

ICU: Intensive Care Unit

MRSA: Methicillin-Resistant *Staphylococcus aureus*

PSI Score: Pneumonia Severity Index Score

PO: per os

IV: intravenous

OD: once daily

BID: bis in die, twice a day

TDS: ter die sumendum, three times a day

QID: quater in die, four times a day

ASSESSMENT OF PATIENTS

Patient history

- Symptoms of an acute lower respiratory tract illness: cough, sputum production, shortness of breath, chest pain.
- Systemic signs: fevers, shivering, sweating
- Co-morbidities: COPD, Asthma, Diabetes Mellitus, Chronic Kidney Disease, Congestive Heart Failure, Coronary Artery disease, malignancy, asplenia, chronic neurological diseases and chronic liver diseases
- Smoking and alcohol abuse
- HIV status
- Recent (last 3 months) antibiotic use
- Recent hospitalization (last 3 months)
- Tuberculosis infection / treatment in the past

Physical examination

- Vital signs: Respiratory rate, oxygen saturation, temperature, blood pressure, heart rate
- Inspection, palpation, percussion and auscultation of chest

Etiology

Organisms commonly associated with pneumonia are listed below.

Bacterial pathogens:

- *Streptococcus pneumoniae*, *Moraxella catarrhalis*, *Staphylococcus aureus*, *Haemophilus influenzae*, *Chlamydia pneumoniae*, Group A *Streptococci*, Enterobacterales.
- Atypical non-zoonotic bacterial pathogens: *Mycoplasma pneumoniae*, *Legionella* spp., *Chlamydia pneumoniae*
- Atypical zoonotic bacterial pathogens: *Chlamydophila psittaci* (psittacosis), *Coxiella burnetii* (Q fever).
- Bronchiectasis or CF: *Pseudomonas aeruginosa*
- *Mycobacterium tuberculosis* can also present as CAP

Viral pathogens:

- Influenza, SARS-CoV-2, other coronaviruses, rhinoviruses, parainfluenza viruses, adenoviruses, respiratory syncytial virus, human metapneumovirus.

Pathogens related to CAP in immunocompromised patients are *Burkholderia*, fungi, *P. jirovecii*, *Nocardia* species.

Diagnostics

The following tests are recommended:

- Chest X-ray (CXR) is recommended as primary radiographic imaging.

Expected findings on chest imaging:

- Lobar consolidation
- Cavitory lesions
- Interstitial pattern
- In cases with high suspicion and inconclusive CXR, an additional chest CT is suggested.

Compare with previous images if available.

- Full blood count
- C-reactive protein
- Procalcitonin in patients who need admission.
- Urea, creatinine and electrolytes as part of severity assessment
- Arterial blood gasses in patients with severe pneumonia
- COVID-19 rapid diagnostic test or PCR when findings are suggestive of viral pneumonia
- Blood culture (2 bottles for adults) in patients who need admission
- Sputum culture can be valuable in certain settings, including in hospitalized patients with severe symptoms, and when patients have a history of MRSA, *Pseudomonas* or Enterobacterales. It is not recommended as a routine investigation for CAP in the outpatient setting.
- HIV screening in severe pneumonia patients
- In cases of severe pneumonia, rapid tests for *Streptococcus pneumoniae* and *Legionella*, along with respiratory virus panel PCR tests (including influenza), can be requested based on their availability both within KFH and in reference laboratories.

DECISION TO HOSPITALISE PATIENTS

Not all patients with pneumonia require hospital admission. A risk-assessment should be performed for all patients with CAP to decide which patient can safely be managed at home. Main decision is based on severity of the disease. Other important considerations are the patient's socio-economic status and home circumstances.

SEVERITY OF ILLNESS ASSESSMENT

A number of severity of illness assessment tools have been developed to assist clinicians in assessing the severity of community-acquired pneumonia. Assessment of the severity of infection is important since it will determine appropriate site of care, microbiological work-up and empiric antibiotic treatment. The Infectious Disease Society of America (IDSA) recommended in 2019 using the PSI score (Table 1) over CURB-65 score (Table 2) to determine which patient can be safely treated as an outpatient

Table 1 - Pneumonia Severity Index (PSI)

Patient characteristics	Number of points
Demographic factors	
Age	
Men	Age in years
Women	Age in years-10
Nursing home resident	Age plus ten
Coexisting illnesses (definitions listed below)	
Neoplastic disease	30
Liver disease	20
Congestive heart failure	10
Cerebrovascular disease	10
Renal disease	10
Physical examination findings	
Altered mental status	20
Respiratory rate >30/min	20
Systolic blood pressure <90 mmHg	20
Temperature <35°C (95°F) or >40°C (104°F)	15
Pulse rate >125/min	10
Laboratory and roentgenographic findings	
Arterial pH <7.35	30
Blood urea nitrogen >30 mg/dL (11 mmol/L)	20
Sodium <130 mmol/L	20
Glucose >250 mg/dL (14 mmol/L)	10
Hematocrit <30%	10
Partial pressure of arterial oxygen <60 mmHg	10
Pleural effusion	10

Risk class	Criteria
I	Age <50 years No existing illnesses or vital sign abnormalities
II	<70 points
III	71 – 90 points
IV	91 – 130 points
V	>131 points

Patients with CAP and scores of <50 (class 1) and 51-70 (class 2) are thought to be mild and potentially suitable for management at home. Patients with scores of 71-90 (class 3) are considered moderately ill and need to be observed in hospital initially. Patients with scores of > 91 (class 4 & 5) are thought to be severely ill, and need consideration for admission to a high-care or even intensive care unit.

Other scores, such as CURB-65, are less accurate, but could be used because of its simplicity and do not require the measurement of the urea level. None of the scoring systems replace clinical assessment. Important additional factors, such as socioeconomic deprivation and co-morbidity, must influence the decision on hospitalization and treatment.

Table 2 – CURB-65

Variable	Value
Confusion	Mental Test Score ≤ 8 , new disorientation in person, place or time
Urea	>7 mmol/L
Respiratory rate	≥ 30 /min
Blood pressure	Systolic < 90 mmHg, and/or diastolic ≤ 60 mmHg
Age	≥ 65 years

The PES (Pseudomonas aeruginosa, extended-spectrum β -lactamase-producing Enterobacteriaceae, and methicillin-resistant Staphylococcus aureus) score (Table 3) was developed in 2015 to predict the microbiological etiology of CAP caused by PES pathogens.

Patients are at risk for MRSA, multidrug-resistant Enterobacteriaceae and *Pseudomonas* if:

- Hospitalization AND IV antibiotics in the last 90 days
- PES score ≥ 5

Table 3 – PES Score *Risk of multidrug-resistant organisms

Score to PES pathogen	Points
Age <40 years	0
Age 40-65 years	1
Age >65 years	2
Male	1
Previous antibiotic use	2
Chronic respiratory disorder	2
Chronic renal disease	3
At emergency department	
Consciousness impairment or aspiration evidence	2
Fever or shivers	-1

In addition to the tables mentioned above, the ATS/IDSA also give recommendations on identifying patients who should be admitted to ICU for intensive monitoring and treatment. Patients requiring ICU-level care are:

- Patients with hypotension requiring vasopressors or respiratory failure needing mechanical ventilation.
- Patients with **three** or more of the following (minor) criteria, of course assessed on individual base:
 - Blood urea nitrogen ≥ 20 mg per dL (7.14 mmol per L)
 - Confusion or disorientation
 - Core temperature $< 36^{\circ}\text{C}$.
 - Hypotension requiring aggressive fluid resuscitation
 - Multilobar infiltrates
 - Partial pressure of oxygen/fraction of inspired oxygen ratio ≤ 250
 - Platelet count $< 100 \times 10^3$ per μL (100×10^9 per L)
 - Respiratory rate ≥ 30 breaths per minute
 - White blood cell count $< 4,000$ per μL (4.00×10^9 per L) due to infection alone (i.e., not chemotherapy induced)

THERAPY AND FOLLOW-UP

Supportive therapy

Patient is given explanation about the diagnosis, prognosis and the therapy he/she will receive.

Analgesics and rehydration are started when indicated.

Oxygen therapy: up to 5L/min via nasal cannula; 6-10L/min via simple non face mask; more than 10L/min via non-rebreathing face mask or high-flow nasal cannula.

Empirical antibiotic therapy

1. Outpatients that will be followed up as outpatient

A. Treatment options for previously healthy outpatients; no antibiotic use in past 3 months

- Amoxicillin 1 gm PO every 8 hours for 5-7 days or
- Amoxicillin-clavulanic acid 625 mg po every 8 hours or 1 gram PO every 12 hours for 5-7 days
or
- Doxycycline 100mg PO every 12 hours for 7 days

PLUS: Azithromycin 500 mg PO daily for 3 days

B. Outpatients with comorbidities or antibiotic use in past three months

- Amoxicillin-clavulanate 1 g PO BID for 5-7 days + Azithromycin 500 mg PO once daily for 3 days or
- Third generation cephalosporin (such as cefixim 400 mg PO daily or 200 mg PO BID) for 5-7 days + Azithromycin 500 mg PO once daily for 3 days or
- Amoxicillin-clavulanate 1 g PO BID + Clarithromycin 500 mg PO BID for 5-7 days

Penicillin allergy:

- Doxycycline 100 mg PO every 12 hours for 7 days
- Levofloxacin* 750 mg PO OD (5 days) or moxifloxacin* 400 mg PO OD (5-7 days)

*Note: Fluoroquinolones should be avoided in patients with suspected TB

If previous colonization with MRSA or *Pseudomonas* or high risk for multidrug-resistant organisms: empirically cover and total treatment will be 7-14 days

2. Inpatients

Beta-lactam antibiotics as intravenous ceftriaxone or amoxicillin-clavulanic acid for 5-7 days PLUS azithromycin 500 mg oral daily for 3-5 days

In case of penicillin allergy: cephalosporins or respiratory fluoroquinolones (levofloxacin) are options.

In patients with severe pneumonia who have a history of prior antibiotic exposure, piperacillin-tazobactam is occasionally chosen as the initial treatment. This decision is always made in consultation with an infectious diseases specialist or pulmonologist.

Previous colonization with MRSA or pseudomonas in previous year OR high risk based on PES-score.

- MRSA: 3rd generation cephalosporin (7 days) + vancomycin + macrolide
- Pseudomonas: Piperacillin-tazobactam / cefepime (7-14 days) + macrolide
- Enterobacteriaceae: piperacillin-tazobactam / cefepime (7-10 days) + macrolide

The American Thoracic Society (ATS)/Infectious Diseases Society of America (IDSA) guidelines

recommended that patients with CAP should be treated for a minimum of five days. Before stopping

therapy, the patient should be afebrile for 48 to 72 hours, breathing without supplemental oxygen (unless required for pre-existing disease), and have no more than one clinical instability factor (defined as heart rate >100 beats/minute, respiratory rate >24 breaths/minute, and SBP $\leq$ 90 mmHg).

Aspiration pneumonia. The 2019 American Thoracic Society (ATS) and Infectious Diseases Society of America (IDSA) guidelines on CAP and the 2023 ERS/ESICM/ESCMID/ALAT guidelines for the management of severe community-acquired pneumonia guidelines recommend against routinely adding anaerobic coverage for patients with (suspected) aspiration pneumonia.

Corticosteroids are not recommended as adjunctive therapy for treatment of non-severe CAP. However, given the potential beneficial effect on length of hospital stay and 28-day mortality, the use of corticosteroids is now recommended in patients with severe CAP who meet one of the following criteria: mechanical ventilation with PEEP > 5 cm water; high-flow oxygen with a FiO₂ > 50% and PaO₂:FiO₂ ratio < 300; nonrebreathing mask with PaO₂:FiO₂ ratio < 300; PSI class V and the absence of relative contraindication (e.g. history suggesting aspiration, pneumonia caused by influenza).

Patients' follow-up and complications

Clinical response is expected within 48 hours otherwise there is a need to reconsider the therapy.

Shifting from IV to oral takes place when patients improve and can take oral therapy.

Complications are to be looked for and managed (empyema, meningitis, septicemia, shock).

After hospital discharge, routine use of follow-up chest imaging in adults with CAP who are improving after the start of antibiotic treatment is not recommended, unless there are complications.

Patients at high risk of lung cancer should benefit from a chest CT scan 4-8 weeks after discharge.

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REVIEW DATE: OCTOBER 2025	REVIEW DATE: OCTOBER 2025
REVISION NUMBER: #4	NEXT REVIEW DATE: OCTOBER 2028

Administrative Data (attachments)

1. References and Bibliography:

1. Martin-Loeches I, Torres A, Nagavci B, Aliberti S, Antonelli M, Bassetti M, Bos LD, Chalmers JD, Derde L, de Waele J, Garnacho-Montero J, Kollef M, Luna CM, Menendez R, Niederman MS, Ponomarev D, Restrepo MI, Rigau D, Schultz MJ, Weiss E, Welte T, Wunderink R. ERS/ESICM/ESCMID/ALAT guidelines for the management of severe community-acquired pneumonia. *Intensive Care Med.* 2023 Jun;49(6):615-632. Erratum in: *Intensive Care Med.* 2023 Aug;49(8):1040-1041.
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2. Revision Contributors Involvement Form

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3. Standards Statements

An audit will be conducted to assess if guidelines are followed. Data in the tables attached will be checked part of the Audit. The expected is 100% compliance

How will monitoring be carried out?	Clinical audit in medical ward and A&E
When will monitoring be carried out?	6 months after starting implementation
Who will monitor compliance with the guideline?	The head of department. AMS Committee

4. Distribution list/dissemination method

Once this guideline has been peer reviewed / consulted upon by stakeholders and has been approved, it will be disseminated to the following:

NO.	SERVICE
1.	Urusaro
2.	Medical
3.	ICU
4.	OPD
5.	Accident & Emergency

6. Clinical Audit tool for Community Acquired Pneumonia

An audit will be conducted to assess if guidelines are followed. Data in the tables below will be assessed as part of the audit. Expected compliance is 100%

1. History taking

Following described	Yes	No	Observations
Respiratory symptoms			
Fever			
Comorbidities			
Recent antibiotic use			

2. Clinical examination records

Following results described	Yes	No	Observations
Oxygen saturation			
Temperature			
Respiratory rate			
Chest examination			

3. Investigations

Following done	Yes	No	Observations
Chest X-ray			
Full blood count			
C-reactive protein or procalcitonin			

4. Treatment

	Yes	No	Observations
Antibiotics prescribed according guidelines			
Duration antibiotics according guidelines			